Cross-Modal Visible-to-Infrared Image Translation in Remote Sensing Guided by Thermal Features

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***Abstract*— Weakly supervised visible (VI)-to-infrared (IR) image translation in remote sensing scenes is a challenging task that attempts to generate corresponding IR images from VI images with cross-model manner, thus tackling the acute scarcity of IR data in certain crucial task scenes. Existing methods frequently concentrate solely on style transfer, resulting in stylistic similarity but physical inconsistency, which limits the authenticity of cross-modal translation. IR thermal features stemming from brightness and darkness contrasts among distinct regions due to temperature disparities, are associated with visual representations in foreground objects in VI images. Therefore, this article proposes ThermalMask, a framework for IR image generation guided by thermal features, aiming to make the generated images more closely with IR features in remote sensing. Specifically, a saliency mask generation network (SMGN) and a semantic attention generation network (SAGN) are designed in the generator. The SMGN is used to generate background and foreground masks to preserve the background and most prominent features of the input remote sensing image. Meanwhile, the SAGN generates attention maps through pyramid nonlocal attention mechanism, guiding the network to adaptively focus on salient regions within both VI and IR feature maps. In addition, to refine the generated IR features, a pixel-spectrum multispatial constraint (PSC) targeting high-frequency components is designed, enabling the generative network to focus more on expressing IR features. Extensive experimental results have shown that our method has good performance in VI-to-IR image translation within remote sensing scenes, surpassing existing state-of-the-art image translation methods. Moreover, it holds promise for application in downstream tasks where IR data is insufficient.**

***Index Terms*— Cross-modal, image translation, thermal features, visible (VI)-to-infrared (IR).**

I. Introduction

RECENTLY, the rapid advancement of remote sensing technology has greatly enhanced the functions of optical cameras on mobile platforms, enabling them to capture a wide range of signals from visible (VI) to infrared (IR) and other multispectral ranges. VI images, characterized by the high resolution and fine texture details, provide intuitive visual representations for scene analysis and object recognition. In contrast, IR images reflect the thermal radiation emitted by objects and backgrounds, and are unconstrained by lighting conditions, exhibiting excellent imaging performance in complex environments [1], [2], [3]. However, the acquisition of IR images faces many challenges such as high imaging costs, restrictions on working environment, and weather conditions, which lead to the scarcity of high-quality IR data resources, making it hard to provide sufficient and diverse learning data for downstream models [4]. The current research is increasingly focusing on the cross-modal generation technology of VI-to-IR images, aiming to learn the potential mapping relationship and conversion mechanism between two modal data. Translating VI images into corresponding IR can provide more data for tasks like object detection [5], personnel rescue [6], and environmental monitoring [7] at low cost in remote sensing.

Currently, there are diverse solutions for achieving cross-modal generation technology that converts VI to IR images. Traditional methods often rely on physical models to artificially simulate thermal regions within images. Qi et al. [8] created a 3-D scene using modeling software and subsequently generated an IR scene through a physical-based approach utilizing reverse path tracking. However, such methods not only exhibit inefficiency but also exhibit significant limitations in terms of accuracy [9]. In view of these challenges, image-to-image intelligent translation methods have gradually emerged as a popular choice. The core of these methods lies in mapping VI images to a latent feature space and reconstructing corresponding IR images through complex nonlinear transformations [10]. Specifically, the introduction of advanced image generation frameworks such as generative adversarial networks (GANs) [11], [12], [13], and diffusion models [14], [15], provides a novel approach for VI-to-IR image conversion. Although the output results of these methods have some similarity to real IR images, they often focus on the transformation of the global structure and style of the image, while neglecting the refined representation of IR thermal features. Furthermore, some methods introduce the concept of contrastive learning into image generation frameworks [16], [17], [18], [19], enhancing the preservation and transformation of detailed information by comparing differences between cross-modal feature blocks. Nevertheless, due to the inherent semantic differences between VI and IR images, these methods still lack additional effective constraints in practical applications. In summary, current cross-modal translation methods mainly rely on pixel-level style transfer networks. Although capable of generating visually plausible IR-like images, these methods merely generate similarly

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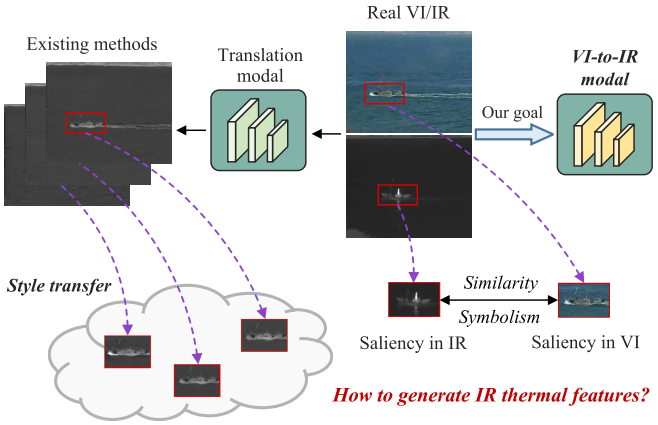


Fig. 1. Illustration of our motivation. The existing methods convert VI images into a style similar to IR, whereas our goal is to explore a new generative model from VI to IR images by leveraging the similarity and symbolism between cross-modal salient features.

stylistic patterns while neglecting the physical principles reflecting thermal radiation characteristics at the feature level during IR imaging. This results in a fundamental discrepancy where generated images exhibit stylistic similarity but physical inconsistency [20].

To enhance the realism of the features in generated remote sensing IR images, this article introduces a novel method, aiming to explore the reconstruction and representation between IR thermal features and VI visual features, as shown in Fig. 1. Our motivation stems from the urgent need for physical authenticity in IR data generation, hoping to highlight the thermal feature regions in the generated IR data. Prior studies suggest that VI and IR modalities share transferable semantic correlations in saliency features, and there is a symbolic relationship between their cross-modal saliency features [21]. The saliency regions in VI images often correspond to potential heat sources, while IR saliency is directly determined by temperature gradients. By training and establishing a mapping relationship between VI saliency and IR thermal features, it is possible to guide the generation model to focus on feature regions that conform to the rules of IR imaging. Therefore, we are dedicated to utilizing these salient features to provide more precise constraint and refinement for IR image generation, thereby improving the distortion issue of existing methods when generating thermal features. Noting that the data collected in real-world scenarios are often unaligned, exhibiting differences in perspective or resolution, so we need to consider the design of networks under weakly supervised conditions [22]. Furthermore, previous researches have demonstrated the importance of frequency domain information in generating authentic content [23], [24], [25]. We posit that the idea of imposing constraints on the pixel space and spectral space of frequency information will be beneficial for the VI-to-IR image translation task in remote sensing.

In this article, we propose ThermalMask, a solution to the problem "stylistic similarity but physical inconsistency" in IR image generation, which utilizes the saliency features of VI modality as a prior for IR thermal feature generation, thereby guiding the learning process of the translation network. By constructing a cross-modal saliency feature mapping network, potential heat source regions can be located and semantically extracted, and detailed information such as edges of these thermal regions can be refined. Specifically, it is an IR image generation network that focuses on salient features. It consists of two generators and two discriminators, forming a dual-branch learning path for VI and IR images. In the generator, a saliency mask generation network (SMGN) and a semantic attention generation network (SAGN) are designed. The SMGN is used to generate foreground masks and background masks, so that both salient foreground features and salient background features in the image can be noticed. The SAGN is used to generate attention maps and combine them with foreground masks to form foreground feature maps that contain specific salient features. To maintain background consistency, the background mask is combined with the source domain to obtain a salient background feature map. Then, the foreground feature map is fused with the background feature map to generate the result. Furthermore, additional constraint schemes help to learn common features between modalities and independent features within modalities from adversarial training. Therefore, we designed a novel, dedicated constraint (PSC) for IR generation that integrates pixel space and frequency spectrum space. This constraint ensures that the generation network is more attuned to the expression of IR features.

Based on the above design, the model proposed in this article can overcome the bias caused by the direct application of image translation methods to IR data, specifically the discrepancy between the generated features and optical laws caused by pixel-to-pixel conversion. In summary, the contributions of this article are as follows.

1) This work introduces ThermalMask, a new paradigm of VI-to-IR image translation network in remote sensing that utilizes the correlation between latent thermal feature regions and saliency to guide IR image generation. Breaking through the limitations of global uniform transfer in existing methods, the generated results are closer to real IR features.
2) Within the generator, an SMGN and SAGN are designed. They are used to focus on salient foreground objects and salient background features in cross-modal images.
3) A new constraint for refining the details of the generated features is proposed, which combines the pixel space and spectral space of high-frequency components, significantly enhancing the expression ability of IR features, and also effectively collaborating to reconstruct global and key structures.

II. Related Work

A. Image-to-Image Translation

Image-to-image translation aims to transform images from a source domain to a target domain, focusing mainly on style transferring through latent spatial operations [26], [27], [28], [29], [30]. These methods have achieved good results in preserving the structural features of the source domain [31].

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For instance, CUT [16] uses patch comparison loss, CycleGAN [12] uses cycle consistency loss, and feature consistency loss. However, the local receptive field of convolution operations in these methods limits the ability to capture details when dealing with high-resolution images, resulting in artifacts in the generated images. Although the cross-granularity consistency proposed by Fu et al. [33] enhances the coordination between global and local features, it still fails to solve the problem of preserving details at high resolutions. Recent research has begun to explore image generation from the frequency domain, Zhang et al. [34] using discrete wavelet transform to extract high-frequency information, while [23], [24], [25] decomposing images into low-frequency and high-frequency components, and learning image feature information in Fourier spectrum space. These methods effectively enhance the accuracy and detail richness of translation, yet they still fall short in controlling specific attributes or structures of the generated domain. Moreover, image-to-image translation methods based on diffusion model have developed rapidly [35], [36], [37]. Xia et al. [38] offered theoretical justification that it is feasible to transfer the distribution from one domain to another at some intermediate step in a diffusion model. Bai and Fu [39] implemented SAR to optical image translation based on a color supervised diffusion model. Despite the impressive performance of these methods in image generation, they still encounter challenges in controllable translation for specific applications.

B. VI-to-IR Image Translation

Recently, significant progress has been made in cross-modal translation of VI-to-IR images, but numerous challenges still remain to be addressed. The existing methods focus on directly transferring the style of images between two modalities [40], [41], [42], [43]. However, these methods frequently overlook the deep correlation between thermal feature distribution in IR images and texture or color information in VI images. To more accurately capture this cross-modal feature correspondence, recent studies have explored the utilization of deep learning models to fit the mapping correlation between VI and IR images, such as using GANs and their variants to learn nonlinear mapping functions from VI to IR [44], [45]. These models typically rely on a large amount of paired data for training, which limits their potential applications in unpaired or sparse data scenarios. In practical situations, the efficacy of generating IR images is influenced by many external factors, such as different weather conditions during data collection or different angles and wavelengths of imaging equipment. Therefore, further research is needed on the generalization ability of many deep learning-based methods in terms of translation performance.

To address this limitation, some studies have explored unsupervised or weakly supervised learning approaches that utilize potential similarities in unpaired data, such as structural resemblance and edge consistency, to guide the cross-modal translation process from VI to IR. Lee et al. [46] guide the preservation of structural details of VI images by potentially preserving image details. Liu et al. [47] and Zhou et al. [48] attempt to integrate thermal information into the model and then use segmentation and masking operations to highlight the object feature area. The difficulty of this method is how to effectively utilize the segmented or simulated hot regions. Furthermore, given the particularity of IR thermal information, a crucial question is how to effectively encode, transmit, and decode these thermal features to accurately represent the thermal radiation state of the object while preserving the authenticity of the image [49]. One possible approach is to combine deep networks with physical models to simulate the imaging process of thermal radiation, while deep networks are responsible for learning high-level feature representations and image synthesis [50].

C. Cross-Modal Remote Sensing Data Conversion

Due to the complexity of remote sensing scenes and the variability of sensors, converting the observation data from one modality to another presents significant challenges, such as VI-to-IR image translation, radar-to-optical image conversion, and panchromatic-to-multispectral conversion [51], [52]. This emerging task has garnered increasing attention from researchers. Wu et al. [53] proposed a multidomain constrained translation network for the change detection task, capable of generating realistic images in the target domain. Li et al. [54] introduced a cross-domain image conversion method semantic segmentation task in remote sensing. Venkatesan et al. [55] proposed an edge-aware GAN for remote sensing image-to-image translation while maintaining stereo constraints. Furthermore, some studies have attempted to address the cross-domain adaptation issue in remote sensing data through image translation, enhancing the learning capability in remote sensing scenes with limited samples [56], [57], [58]. Han et al. [59] introduced disentangled representation learning to disentangle the image representation of aerial VI and IR images into a semantic structure space and two imaging style spaces. Recently, the remote sensing community has witnessed the emergence of advanced foundation models trained on large-scale multimodal datasets, exhibiting remarkable adaptability in many downstream tasks. SpectralGPT [60] is designed for hyperspectral data processing, incorporating a 3-D generative transformer network built upon a masked autoencoder framework, balancing spatial–spectral coupling encoding and spectral sequence preservation. Cross-city matters [61] achieves domain adaptation across city scenes through adversarial training, and its dynamic feature alignment mechanism effectively solves the problem of geographic distribution offset. These foundation models can provide new solutions for VI-to-IR image translation tasks and provide deep image semantic features for network training. We can use the feature extraction capabilities of pretrained architectures to guide the generation of physically consistent IR thermal patterns.

III. Methodology

This section introduces the proposed aerial VI-to-IR image translation method. Section III-B introduces the overall architecture. Section III-C describes the IR generation guided

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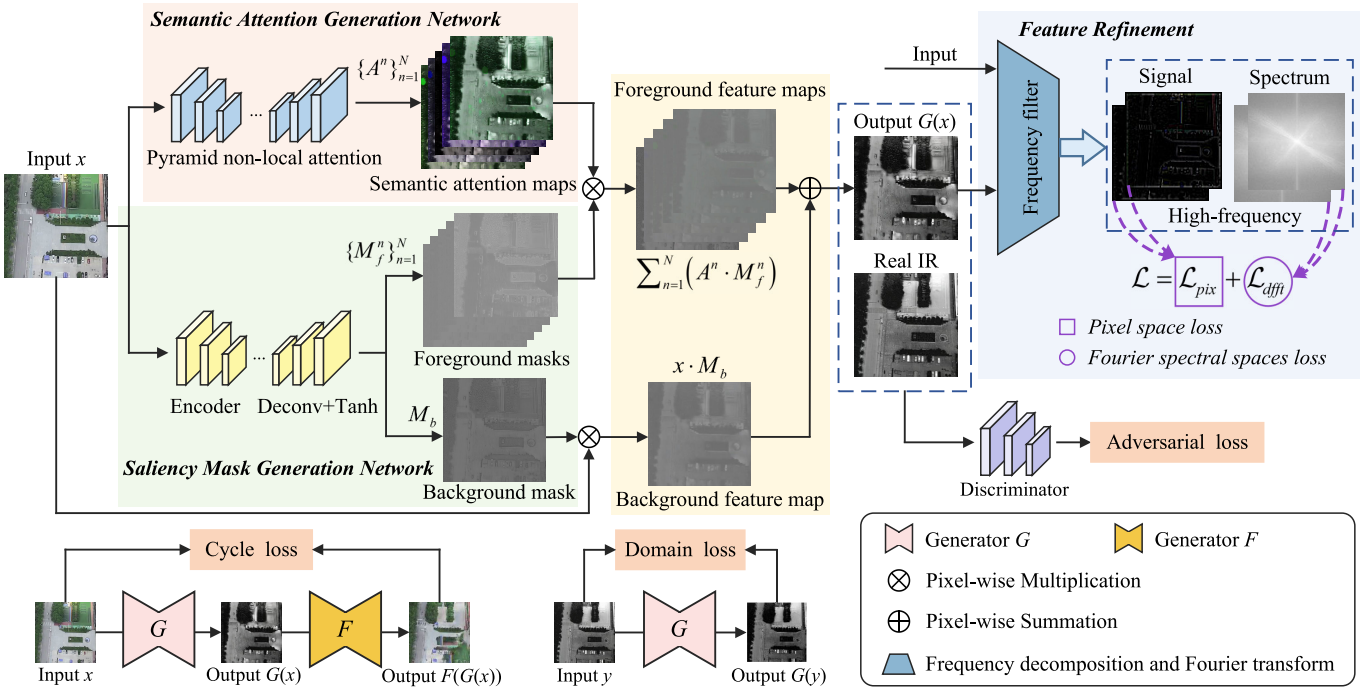


Fig. 2. Overall framework of ThermalMask. We designed SAGN and SMGN modules. The SAGN is used to generate semantic attention maps, while the SMGN is responsible for generating foreground masks and background masks. Moreover, we also designed a feature refinement stage that performs pixel-spectrum multispace constraint.

by thermal features. Next, Section III-D introduces the designed constraints in pixel space and frequency spectral space. Finally, Section IV describes the loss function.

A. Network Architecture

Given a dataset of VI modality $\{X\} \in \mathbb{R}^{H\times W\times 3}$ and an approximately aligned dataset of IR modality $\{Y\} \in \mathbb{R}^{H\times W\times 1}$, including VI images $x \in X$ and IR images $y \in Y$. Our task is to learn the salient feature mapping relationship between two modalities, as well as the constraint relationship of global feature refinement, and to generate images $\tilde{y}$ that conform to IR thermal features. As shown in Fig. 2, the proposed method consists of two generators G and F, and two discriminators D_X and D_y. Each generator contains an SMGN and an SAGN, which focuses on the most significant differences between the source domain and the target domain. Each discriminator performs adversarial learning between the generated image and the corresponding input image. In order to refine the generated IR images from global range, a contrastive learning approach was proposed by comparing the pixel and spectral domains of high-frequency component in VI data and IR data. The whole translation process is denoted as follows:

$$\tilde{y} = f_{X\to Y}(x, y, G(\mathrm{SMGN}(X), \mathrm{SAGN}(X)), D(Y)) \quad (1)$$

where $G(\mathrm{SMGN}(\cdot), \mathrm{SAGN}(\cdot))$ represents the generator network structure, it is composed of SMGN and SAGN networks.

During network training, input the VI image x into the generator G, and the generated IR-like image $\tilde{y}$ is obtained, and multidomain features are learned to constrain. Then, the generated IR-like image $\tilde{y}$ is input into the generator F to obtain the reconstructed VI image $\tilde{x}$, forming a recurrent network. Finally, input the unaligned IR images, exchange the generation and discrimination relationships between IR and VI modalities, form a reverse mapping relationship, and perform cross-modality adversarial constraints. During model inference, only VI images x are input for the network's forward propagation prediction process, generating IR images $\tilde{y}$. Note that the network does not learn pixelwise correspondences, but rather learns the mapping relationship between the two modalities. Therefore, this network architecture is effective for unaligned data and represents a form of weakly supervised learning approach.

B. Thermal Feature-Guided Generation

According to the blackbody radiation theory, thermal radiation is an inherent property of all region, including the background area or foreground objects in an image. However, the thermal region of an image defined in this work specifically refers to the foreground object area that has a significant temperature difference from the surrounding background in the scene. Although RGB sensors are incapable of directly measuring temperature, they can correlate and infer thermal information from visual cues. For instance, in most cases, foreground objects or thermal sources exhibit elevated temperatures than the surrounding background. Consequently, this work uses foreground object areas in VI images as potential thermal feature regions with temperature differences, providing approximate thermal priors for the generation network. This approximate partitioning approach relies on the key assumption that local saliency can approximate relative thermal feature contrasts. To extract feature knowledge useful for cross-modal translation, the proposed method is designed to preserve and map features based on thermal regions. The generator fully considers the saliency expression of

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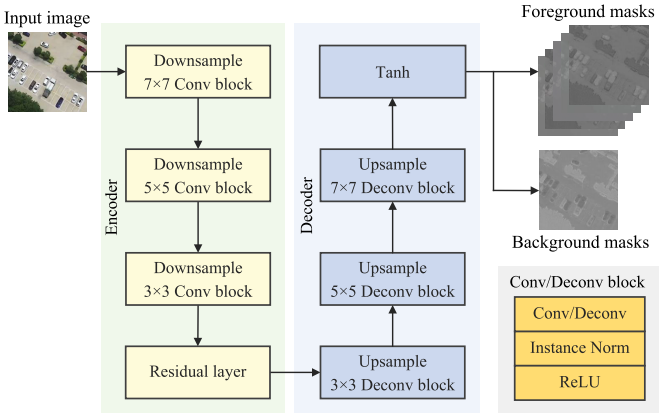


Fig. 3. Proposed SMGN module. In VI images, salient features are typically foreground objects, which correspond to thermal features in IR images. Therefore, the learning process of the generator is guided by the regions where thermal features are located.

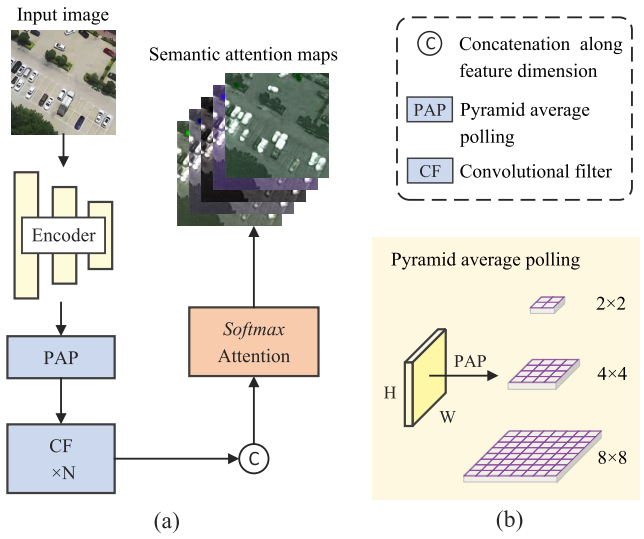


Fig. 4. Proposed SAGN module. The SAGN can obtain a set of semantic attention maps through pyramid nonlocal attention, adaptively focusing on key regions in VI and IR feature maps.

thermal information when generating IR images, while the discriminator pays more attention to the consistency of these saliency regions. By employing this thermal information-guided approach for salient feature mapping, VI images can be translated into images that retain key semantic information and conform to IR characteristics.

In the generator, a salient mask generation network and an SAGN are designed to simultaneously generate masks and compute attention for the input VI images, as shown in Figs. 3 and 4. The SMGN is capable of generating foreground masks and background masks separately, for example, treating objects such as vehicles, airplanes, and ships as salient foregrounds, while other areas are considered as the background. In this way, regions in the image that may possess IR thermal characteristics are distinguished from the background. Specifically, the SMGN extracts image feature maps m through convolution layers and generates N foreground masks $\{M_f^n\}_{n=1}^N$ and one complementary background mask M_b through deconvolution blocks in the deeper layers of the network. The process is as follows:

$$\left\{M_f^n\right\} = \text{Deconv}\left(W_f^n \cdot m + b_f^n\right) \tag{2}$$

where W_f^n is the weight of the nth foreground mask filter, b_f^n is the bias term, and Deconv represents the deconvolution operation, $n = 1, \ldots, N$. Generating multilayers of foreground masks is to fully explore salient feature knowledge. If only the generated mask layer is one, it may lead to incomplete mining of object information. By using SMGN, salient features from VI and IR training data can be captured, which actually contain possible thermal information in the image.

The SAGN can obtain a set of semantic attention maps $\{A^n\}_{n=1}^N$ by utilizing pyramid nonlocal attention [62], [63], enhancing the network's ability to capture contextual information and adaptively focus on key regions in VI and IR feature maps. A pyramid nonlocal attention module is introduced into the network to capture long-range dependencies, effectively utilizing multiscale features within the pyramid. Specifically, average pooling operations of different sizes are applied to the input feature map to generate a series of multiscale features, which are further extracted through N convolution filters and upsampled to the original spatial dimensions. Finally, the Softmax weights are computed. In this way, the three-channel generation space can be expanded to a $3(N + 1)$-channel generation space, which is more suitable for learning effective mappings of cross-modal features. The process is as follows:

$$\{A^n\} = \text{Softmax}\left(W_a^n \cdot t + b_a^n\right) \tag{3}$$

where W_a^n denotes the weight of nth convolution filter, b_a^n is the bias term, and the Softmax represents the pyramid nonlocal attention computation, $n = 1, \ldots, N$. Subsequently, the salient foreground mask $\{M_f^n\}_{n=1}^N$ is elementwise multiplied with the semantic attention map $\{A^n\}_{n=1}^N$ to retain prominent foreground objects while filtering out the background from the VI domain. Meanwhile, the salient background mask M_b is elementwise multiplied with the input image x of the source domain to generate a salient background feature map. Through the dual-branch multiplication operation, the generator G acquires the ability to learn and translate salient features. Finally, the foreground feature map and the background feature map are elementwise added to obtain the target domain image generated from the source domain image. Formally, it can be written as follows:

$$G(x) = x \cdot M_b + \sum_{n=1}^{N} \left(A^n \cdot M_f^n\right). \tag{4}$$

Through the above manner, the network is able to retain the background knowledge of the source domain while focusing on the foreground features of the source domain. The structure of generator F is similar to that of generator G, consisting of an SMGN and an SAGN. During the learning process of the two generators G and F, salient information such as textures and edges from VI data, as well as thermal characteristics from IR data, are effectively utilized to provide feature knowledge guidance in the generative network.

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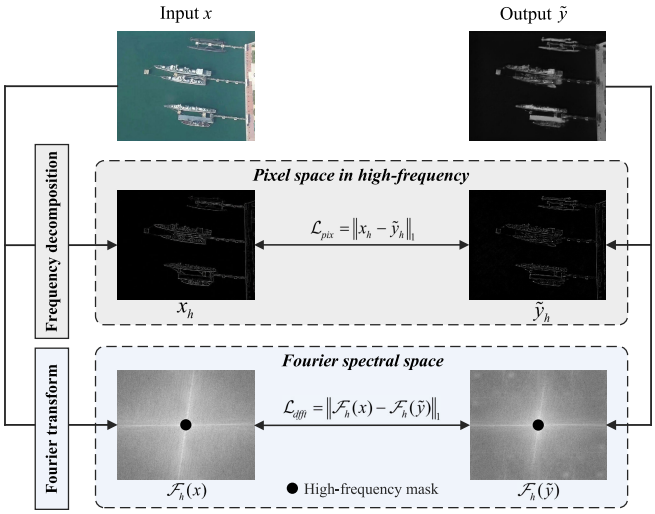


Fig. 5. Proposed feature refinement constraints. Constraining both the pixel space and frequency spectrum space of high-frequency information for refining IR thermal features.

C. Cross-Modal Feature Refinement Constraint

There is a strong correlation between the thermal features captured by real IR sensors and the salient information in images, yet differences also exist. Therefore, the generation guided by salient features is considered as the first stage of the entire translation process, and the results at this stage are relatively coarse, so the task of the next stage is to refine the feature representation of the generated image to better match the thermal feature distribution of the IR images. Previous research has demonstrated the importance of frequency domain information for the authenticity of generated content. To optimize image details both globally and locally, this section designs feature refinement loss based on pixel-spectrum multispatial constraint (PSC), aiming to make the generated IR feature representation more realistic, as shown in Fig. 5.

Specifically, a Gaussian kernel is first used as a frequency domain filter to decompose the source VI image X and the translated image $\tilde{y}$ obtained in the first stage into their respective high-frequency components x_h, $\tilde{y}_h$ and low-frequency components x_l, $\tilde{y}_l$. High-frequency components highlight the edges and details of the image, while low-frequency components emphasize the color and overall structure. Since high-frequency and low-frequency features are complementary in the composition of frequency domain, we only design content difference minimization constraint on the high-frequency components x_h and $\tilde{y}_h$ in the pixel space of the frequency domain feature maps. Note that if the content difference constraint is also imposed on the low-frequency components x_l and $\tilde{y}_l$, it can lead to overly smooth generation results, with important details and texture information lost. Therefore, the pixel space loss $\mathcal{L}_{\text{pix}}$ for the high-frequency feature maps is calculated as follows:

$$\mathcal{L}_{\text{pix}} = \mathbb{E}_{x\sim X,\tilde{y}\sim\tilde{Y}}\left[\left\|x_h - \tilde{y}_h\right\|_1\right]. \tag{5}$$

Moreover, we expect that the features generated in this section have a similar spectral space to those of the source image. The similarity in the spectral space actually reflects the consistency in frequency components and energy distribution of the time domain signals, which is crucial for the structural consistency of the generated features. Therefore, we design constraints on the consistency of the Fourier spectra between the source VI image and the generated IR image. First, the discrete Fourier transform (DFT) is applied to both the source image and the generated image to obtain their signals $\mathcal{F}(x)$ and $\mathcal{F}(\tilde{y})$ in the Fourier spectral space. The formula for calculating the DFT of an image of size $H \times W$ is as follows:

$$\mathcal{F}(V)(u,v) = \frac{1}{H\cdot W}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} e^{-2\pi j\left(\frac{h\cdot u}{H}+\frac{w\cdot v}{W}\right)}\cdot V(h,w) \tag{6}$$

where $\mathcal{F}(u,v)$ is the index in the spectral domain, $u = 0,\ldots,H-1$ and $v = 0,\ldots,W-1$. $V(h,w)$ is the pixel index of the image in the frequency spatial domain. Each signal point in the Fourier spectrum utilizes information from the pixel space to represent the global-level frequency domain feature distribution. Subsequently, a high-frequency mask is applied to retain the high-frequency information in the spectral signals, and the high-frequency signals $\mathcal{F}_h(x)$ and $\mathcal{F}_h(\tilde{y})$ in the two Fourier spectral spaces are calculated as follows:

$$\begin{aligned}\mathcal{F}_h(x) &= \mathcal{F}(x)\odot M_H\\ \mathcal{F}_h(\tilde{y}) &= \mathcal{F}(\tilde{y})\odot M_H\end{aligned} \tag{7}$$

where M_H is the high-frequency signal mask, used to selectively preserve high-frequency information, $\odot$ representing the exclusive NOR operation. The definition of high-frequency signal mask is as follows:

$$M_H(u,v) = \begin{cases}1, & (u-u_c)^2+(v-v_c)^2 \geqslant r^2\\ 0, & \text{other}\end{cases} \tag{8}$$

where u_c and v_c are the center coordinates of the spectrum, and r is the radius parameters of the mask range. When the distance between the coordinates of a point in the spectrum and the center coordinates is greater than or equal to r, the information of that point is retained. After that, the consistency of the high-frequency signals $\mathcal{F}_h(x)$ and $\mathcal{F}_h(\tilde{y})$ in the two Fourier spectrum spaces is constrained, and the loss $\mathcal{L}_{\text{dfft}}$ is calculated as follows:

$$\mathcal{L}_{\text{dfft}} = \mathbb{E}_{x\sim X,\tilde{y}\sim\tilde{Y}}\left[\left\|\mathcal{F}_h(x) - \mathcal{F}_h(\tilde{y})\right\|_1\right]. \tag{9}$$

The feature refinement based on the pixel-spectrum constraint that targets high-frequency information effectively enhances the IR feature expression in translated images and improves the efficacy of salient feature generation.

D. Objective Function

1) Cycle Loss Guided by Salient Features: In order to further standardize the mapping, this section adopts a bidirectional cycle loss similar to CycleGAN [12] in the generation process. It ensures consistency in the conversion of IR and VI features in both fields. Specifically, for the generator F, it can restore the output of the generator G to the original state of VI image x, that is: $x \to G(x) \to F(G(x)) \approx x$. Similarly, the generator G can restore the output of the

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generator F to its original state of IR image y. The loss of the entire cycle $\mathcal{L}_{\text{cyc}}$ is as follows:

$$\mathcal{L}_{\text{cyc}} = \mathbb{E}_{x\sim X}\left[\|F(G(x)) - x\|_1\right] + \mathbb{E}_{y\sim Y}\left[\|G(F(y)) - y\|_1\right]. \tag{10}$$

Both the generators G and F performed salient mask and semantic attention calculations, so the process of performing cycle-consistency loss is guided by the salient feature calculation process. Fuse the feature masks obtained from the SMGN with the semantic attention maps obtained from the SAGN to obtain an improved representation of cycle generation. The process is as follows:

$$\begin{aligned} F(G(x)) &= G(x)\cdot \hat{M}_b + \sum_{n=1}^{N}\left(\hat{A}^n \cdot \hat{M}_f^n\right) \\ G(F(y)) &= F(y)\cdot M_b + \sum_{n=1}^{N}\left(A^n \cdot M_f^n\right) \end{aligned} \tag{11}$$

where $\hat{M}_b$, $\hat{M}_f^n$, and $\hat{A}^n$ are respectively the background mask, foreground mask, and semantic attention map generated by the generator F. In the formula, $F(G(x))$ should be very close to VI images x. Similarly, $G(F(y))$ should be very close to real IR images y. The bidirectional cycle loss guided by the salient features designed above can control the generation network to maintain better style consistency for IR reference content and improve the low-frequency feature quality of the generated image.

2) Adversarial Loss: Adversarial loss constrains the similarity between the generated image and the real image through discriminators D_X and D_Y. This loss ensures that the translated IR image maintains its basic structure in the VI domain. Based on the design in GANs [11], the adversarial loss $\mathcal{L}_{\text{adv}}^y$ generated by the discriminator D_Y is calculated as follows:

$$\mathcal{L}_{\text{adv}}^y = \mathbb{E}_{y\sim Y}\left[\log D_Y(y)\right] + \mathbb{E}_{x\sim X}\left[\log\left(1 - D_X(\tilde{y})\right)\right]. \tag{12}$$

Similarly, the loss $\mathcal{L}_{\text{adv}}^x$ generated by the discriminator D_X can be obtained as follows:

$$\mathcal{L}_{\text{adv}}^x = \mathbb{E}_{x\sim X}\left[\log D_X(x)\right] + \mathbb{E}_{y\sim Y}\left[\log\left(1 - D_Y(\tilde{x})\right)\right]. \tag{13}$$

Therefore, the overall adversarial loss $\mathcal{L}_{\text{adv}}$ is the sum of $\mathcal{L}_{\text{adv}}^x$ and $\mathcal{L}_{\text{adv}}^y$.

3) Domain Conversion Loss: We hope that the generator G processes a certain level of domain generalization ability, enabling it to map input features to the IR domain irrespective of whether the input is a typical VI image. In other words, G could convert VI into IR domain image, if the IR image is input, the output should still be an IR domain image, that is, $G(y)$ as similar as possible to the IR image $y \in Y$. Similarly, $F(y)$ should be as similar as possible to the VI images $x \in X$. This constraint can control the overall style of the generated modality, and the domain conversion loss $\mathcal{L}_{\text{dom}}$ is designed as follows:

$$\mathcal{L}_{\text{dom}} = \mathbb{E}_{x\sim X}\left[\|F(x) - x\|_1\right] + \mathbb{E}_{y\sim Y}\left[\|G(y) - y\|_1\right]. \tag{14}$$

4) Total Loss: Using the loss term designed above, we obtain the objective function in network backpropagation. The final loss value is the weighted sum of various loss values, in the following form:

$$\mathcal{L} = \alpha\cdot\mathcal{L}_{\text{cyc}} + \beta\cdot\mathcal{L}_{\text{adv}} + \delta\cdot\mathcal{L}_{\text{dom}} + \eta\cdot\left(\mathcal{L}_{\text{pix}} + \mathcal{L}_{\text{dfft}}\right) \tag{15}$$

where α, β, δ, and η are the weight coefficients that adjust the weights of cycle-consistency loss, adversarial loss, domain conversion loss, and feature refinement loss.

IV. Experiments

A. Dataset and Setup

1) AVIID Dataset [10]: This publicly available dataset is road scene images captured in VI and IR cameras, including objects such as cars, buses, small trucks, and large trucks. It contains three subdatasets, among which AVIID-1 contains 993 pairs of daytime images collected from a road monitoring perspective, with the resolution of 434 × 434; AVIID-2 includes 1070 pairs of nighttime images collected from a similar monitoring angle, also with the resolution of 434 × 434; and AVIID-3 consists of 1280 road vehicle images captured by drones at varying heights, with the resolution of 512 × 512. The scenes depicted in these images are consistent, primarily including various vehicles, roadside trees, and houses.

2) IRVI Dataset [64]: This public dataset is a continuous frame dataset of vehicle driving scenes, primarily comprising vehicles and backgrounds captured from an autonomous driving perspective. To prevent data leakage, different video clips were selected for the training and testing sets. The dataset includes 17 000 pairs of images for training and 1000 pairs for testing, and the image size is 256 × 256.

3) DroneCoast Dataset: Currently, there is a scarcity of publicly available datasets for VI and IR aerial photography of coastal scenes. Therefore, we utilized binocular mid-wave IR and color cameras to collect video streams of coastal scenes and constructed a dataset of mid-wave IR and VI images specifically for drone aerial photography. Video data was obtained through onboard and bracket-mounted cameras, with rough alignment of the scenes in both modalities achieved at the hardware level. The dataset comprises 939 pairs of training images and 164 pairs of test images, each with a resolution of 640 × 512. The desired outcome for this scene is to map the coastal images from the VI domain to the IR domain as closely as possible.

4) Implementation Details: The batch size for training the network in this article was set to 1, and the initial learning rate was specified to 1×10^{-4}. A total of 200 training epochs were employed, with a 50% learning rate decay every 50 epochs. Additionally, loss weights $\alpha = 2$, $\beta = 1$, and $\delta = 1$ were set accordingly. All training and test experiments in this article were conducted on a workstation equipped with an RTX 4090 GPU.

5) Comparison Methods: We compared the proposed method with advanced universal image translation methods: CycleGAN [12], CUT [16], Swin-UNIT [25], PUT [18], HnegSRC [27], DMT [38], and StegoGAN [26], and compared

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TABLE I
ABLATION STUDY OF NETWORK COMPONENTS ON OVERALL PERFORMANCE. BOLD IS THE BEST PERFORMANCE

Setup	SMGN	SAGN	PSC	AVIID-1			IRVI			DroneCoast		
				SSIM↑	PSNR↑	FID↓	SSIM↑	PSNR↑	FID↓	SSIM↑	PSNR↑	FID↓
Baseline				0.716	24.61	58.87	0.562	18.52	110.66	0.682	24.24	170.90
Model-1	✓			0.625	24.56	149.44	0.517	17.26	128.33	0.655	23.36	189.33
Model-2		✓		0.741	26.37	80.60	0.594	19.53	116.71	0.730	24.72	166.25
Model-3			✓	0.780	27.53	48.31	0.610	19.70	100.51	0.743	25.26	156.98
Model-4	✓	✓		0.823	28.05	47.23	0.635	20.17	82.77	0.759	24.91	135.40
ThermalMask	✓	✓	✓	**0.886**	**30.17**	**38.94**	**0.681**	**22.24**	**75.32**	**0.784**	**25.31**	**122.97**

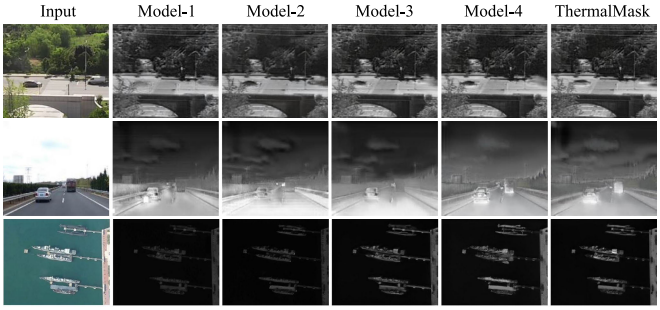


Fig. 6. Results of ablation study on the effectiveness of components in AVIID-1, IRVI, and DroneCoast datasets. Model-1 uses SMGN module; Model-2 uses SAGN module. Model-3 uses PSC loss. Model-4 uses both SMGN and SAGN.

with advanced cross-model image translation methods: RGB-TIR [46] and DR-AVIT [59]. All these methods can perform weakly supervised image-to-image conversion without paired training data. Among them, the universal image translation method can perform IR stylization conversion. RGB-TIR and DR-AVIT are image translation methods specifically designed for generating IR data, which can focus on the thermal features in the training data. We use these models as comparison methods, training and testing them on the three datasets introduced above using their official version settings. During the comparative experiment, we implemented the following four systematic measures to ensure that all comparative methods achieve their maximum attainable performance: hyperparameter replication, performance optimization for low-ranked methods, data consistency, and environmental standardization.

B. Evaluation Indicators

We use four quantitative indicators to evaluate the comparison results between the proposed method and other state-of-the-art methods: structure similarity (SSIM), peak signal-to-noise ratio (PSNR), Fréchet inception distance (FID) [65], and kernel inception distance (KID) [66]. The SSIM is an indicator that measures the similarity between the generated image and the real image. The closer the value is to 1, the more similar the generated image is to the corresponding real image. The PSNR can measure the degree of distortion between the generated image and the corresponding real image. The higher the value, the smaller the degree of distortion in the generated image. The FID is a widely used quality assessment indicator for image generation. It is used to estimate the feature distribution between the real image and the generated image by calculating the depth features extracted by the Inception-V3 model. The KID measures the feature distribution between the generated image and the real image by mapping the feature vectors to a high-dimension space and calculating the difference in the eigenvalues of the Gaussian kernel matrix between them. The smaller the values of FID and KID, the closer the distribution of the two images in the feature space. Furthermore, we also used the precision, recall, and average precision (AP) under different intersection-over-union [67] in object detection to evaluate the performance of the IR images converted by translation methods in downstream detection and recognition tasks.

C. Ablation Analysis

1) Impact of Network Structure: This section studies the impact of network components on overall performance. On the basis of the CycleGAN model in this article, we gradually added the components of the ThermalMask model, that is, SMGN, SAGN, and the PSC. The experiment results on the AVIID-1, IRVI, and DroneCoast datasets are shown in Table I. Among them, Model-1 only introduces SMGN, and since there is no involvement of SAGN, the foreground mask will directly learn the semantic features of the source image. Model-2 only introduces SAGN and does not have salient region generation guided by thermal features. Model-3 only introduces PSC without adding other network components. Model-4 introduces both SMGN and SAGN for the computation of foreground feature maps and background feature maps, without adding the feature refinement process. It can be seen that the lack of any of these three designs will significantly reduce the performance of generating IR images, and the combination of all these modules is crucial for weakly supervised VI-to-IR image conversion learning. The visual effect after adding components is shown in Fig. 6.

Moreover, we observed even worse results when using SMGN alone. The foreground mask generated by SMGN only approximates the salient features in the source image, representing the regions where thermal features may exist. If the semantic information of the source image is not extracted and learned, the foreground feature map cannot be effectively synthesized, resulting in poor generation performance. When only refinement constraints are used without incorporating SMGN and SAGN, it means that the model, based on the original cycle-consistency constraints, adds additional constraints on the generation effect using frequency domain information, thus resulting in better results. When the SMGN

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TABLE II
COMPARISON OF THE CONTRIBUTIONS OF $\mathcal{L}_{\text{pix}}$ AND $\mathcal{L}_{\text{dfft}}$ TO NETWORK PERFORMANCE

Setup	$\mathcal{L}_{pix}$	$\mathcal{L}_{dfft}$	AVIID-1			IRVI			DroneCoast		
			SSIM↑	PSNR↑	FID↓	SSIM↑	PSNR↑	FID↓	SSIM↑	PSNR↑	FID↓
Model-4			0.823	28.05	47.23	0.635	20.17	82.77	0.759	24.91	135.40
Model-5	✓		0.861	29.56	41.25	0.652	21.42	80.69	0.775	25.25	125.59
Model-6		✓	0.857	29.04	44.78	0.668	21.89	77.48	0.767	24.99	128.71
ThermalMask	✓	✓	**0.886**	**30.17**	**38.94**	**0.681**	**22.24**	**75.32**	**0.784**	**25.31**	**122.97**

and SAGN are integrated into the generative network, the attention map is fused with the foreground mask, and the source image is fused with the background mask, allowing the network to fully capture potential thermal regions, making the generated results more realistic and similar to actual IR images.

Furthermore, based on Model-4, we compared the impact of $\mathcal{L}_{\text{pix}}$ and $\mathcal{L}_{\text{dfft}}$ on network performance in Table II. Model-5 introduces $\mathcal{L}_{\text{pix}}$, it will improve the structural similarity and feature similarity results of generated images. Model-6 introduces $\mathcal{L}_{\text{dfft}}$, it will also improve the overall performance of the network. The results indicate that when introducing $\mathcal{L}_{\text{pix}}$ or $\mathcal{L}_{\text{dfft}}$ separately, the performance of the network can be improved to a certain extent, and the contributions of both to the network are comparable. Combining the two constraints into a refinement loss for feature generation can have a positive impact on the network. It is also interesting to note that the contribution of $\mathcal{L}_{\text{pix}}$ to the improvement for results is greater than that of $\mathcal{L}_{\text{dfft}}$ on the AVIID-1 dataset, while the contribution of $\mathcal{L}_{\text{pix}}$ is less than that of $\mathcal{L}_{\text{dfft}}$ on the IRVI dataset. This is because the image features of the AVIID-1 dataset are complex and fine, requiring the model to achieve high restoration at the pixel level, while $\mathcal{L}_{\text{pix}}$ can perform a higher contribution in scenes that require fine reconstruction. The image features of the IRVI dataset emphasize the edges and overall structure of the objects, while $\mathcal{L}_{\text{dfft}}$ can more effectively capture the frequency features of the image, playing a significant role in the optimization process. Therefore, when optimizing the model, it is not sufficient to rely solely on a single loss function; rather, it is important to comprehensively consider the balance between the two types of losses to adapt to the challenges presented by different datasets.

2) Analysis of Hyperparameters: The constraint calculation on generating refined features in the total loss function is crucial for guiding the convergence process of network training, and its loss value directly affects the effectiveness of high-frequency feature constraints. We compared the impact of weight coefficients η controlling the loss value on the performance of ThermalMask on the IRVI dataset, and the results are shown in Table III. With the remaining hyperparameters α, β, and δ following the original settings in the baseline network, we experimented with five different values for hyperparameter η to identify the optimal weights for feature refinement constraints. It can be observed that, in the case of hyperparameters $\eta = 1$, the generation effect of the proposed model is better than other weight values, which verifies the effectiveness of the hyperparameter values used in this article. When $\eta \leqslant 0.5$, the constraints generated by frequency domain information were too small, and the model was unable to effectively capture high-frequency features, resulting in poor quality of the generated images. Conversely, when $\eta \geqslant 5$, the constraints generated by frequency domain information were too large, it had a negative impact on the model training process and also led to a decrease in network performance. Therefore, the hyperparameters η were set to 1 for training.

TABLE III
STUDY ON THE WEIGHT OF HYPERPARAMETER η IN IRVI DATASET

$\eta \cdot (\mathcal{L}_{pix} + \mathcal{L}_{dfft})$	$\eta = 0.1$	$\eta = 0.5$	$\eta = 1$	$\eta = 5$	$\eta = 10$
SSIM ↑	0.642	0.670	**0.681**	0.656	0.613
FID ↓	102.57	82.16	**75.32**	87.52	117.30

TABLE IV
STUDY ON THE NUMBER OF CONVOLUTION FILTERS N FOR THE OVERALL PERFORMANCE IN IRVI DATASET

number	$N = 1$	$N = 5$	$N = 7$	$N = 10$	$N = 15$
SSIM ↑	0.652	0.670	**0.681**	0.675	0.673
FLOPs ↓	67.35	69.43	**70.26**	71.54	73.71

In addition, in the design of SMGN and SAGN, N convolution filters were used to generate foreground masks $\{M_f^n\}_{n=1}^N$ and semantic attention maps $\{A^n\}_{n=1}^N$, respectively. To investigate the specific impact of the number of layers on model performance, we conducted experiments on the IRVI dataset to validate the value of N. When using more layers, the number of parameters and floating-point operations (FLOPs) will increase accordingly, so it is necessary to find the optimal balance between computational efficiency and generation effectiveness. Table IV lists the impact of the number of layers N on the performance of ThermalMask. As can be seen, when $N = 7$, the structural similarity between the generated IR image and the real IR image reached a peak of 0.681, and the FLOPs value was 70.26G, which was within an acceptable and reasonable range. When the number of layers is further increased, such as $N = 10$ and $N = 15$, although the computational complexity of the model continues to rise, the SSIM value does not increase accordingly, and it shows a downward trend instead. This phenomenon may be due to excessive overlap of foreground information in the image, which interferes with the model's accurate capture of image features and thus affects the optimal performance of image generation. When $N = 1$, though the FLOPs value slightly decreases, the SSIM indicator decreases by 2.9%. Therefore, in SAGN and SMGN, setting the number of foreground masks

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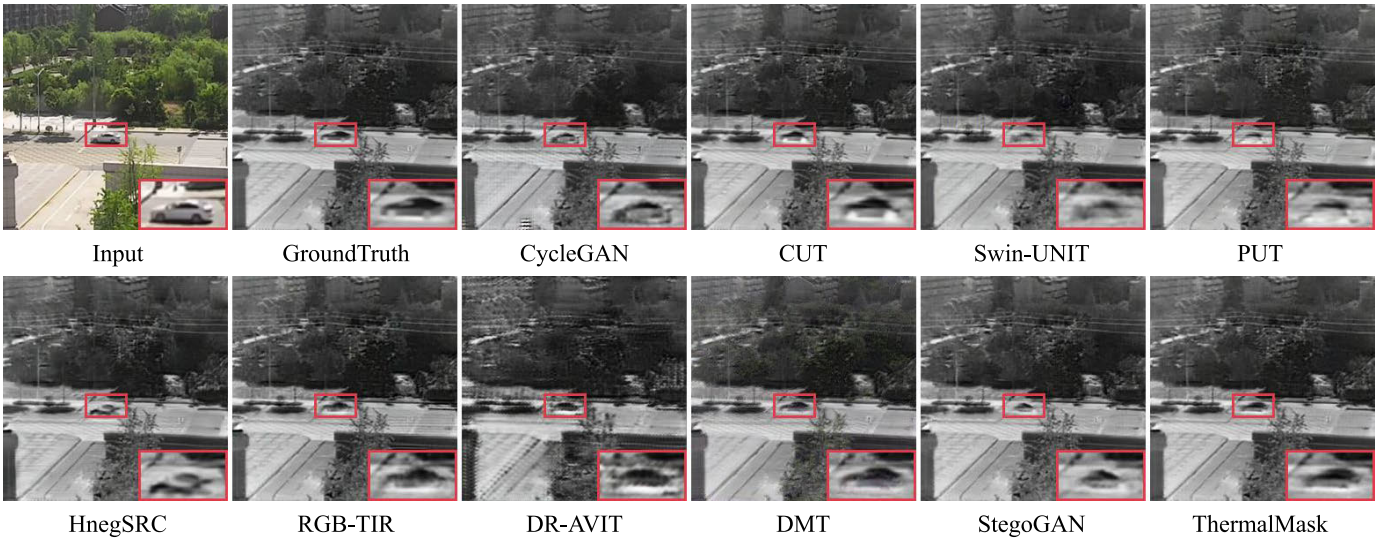


Fig. 7. Qualitative comparisons on the AVIID-1 dataset (daytime scene in drone remote sensing).

TABLE V
QUANTITATIVE COMPARISON ON AVIID DATASET. BOLD AND UNDERLINE DENOTE THE BEST AND SUBOPTIMAL PERFORMANCE, RESPECTIVELY

Public	Method	AVIID-1				AVIID-2				AVIID-3			
		SSIM↑	PSNR↑	FID↓	KID↓	SSIM↑	PSNR↑	FID↓	KID↓	SSIM↑	PSNR↑	FID↓	KID↓
ICCV 17	CycleGAN [12]	0.71	24.61	58.87	0.049	0.71	25.11	100.53	0.128	0.52	16.51	92.80	0.012
ECCV 20	CUT [16]	0.72	24.45	69.55	0.033	0.73	24.16	52.84	0.031	0.53	17.94	**77.73**	**0.009**
MM 23	Swin-UNIT [25]	0.80	25.96	86.65	0.097	0.65	19.15	113.92	0.146	0.52	15.88	93.47	0.013
MM 22	PUT [18]	0.70	24.14	37.95	**0.023**	0.58	20.90	85.13	0.106	0.51	17.19	91.98	0.014
CVPR 22	HnegSRC [27]	0.59	22.43	53.64	0.043	0.66	23.00	46.78	0.032	0.47	18.00	96.51	0.031
ICRA 23	RGB-TIR [46]	0.81	25.59	**36.80**	0.028	0.80	25.76	50.29	0.031	0.48	15.18	93.71	0.016
TGRS 24	DR-AVIT [59]	0.55	22.08	142.36	0.174	0.53	16.12	209.38	0.319	0.42	14.33	154.02	0.078
TPAMI 24	DMT [38]	0.75	26.58	118.89	0.171	0.64	21.31	111.01	0.141	0.55	18.98	155.82	0.109
CVPR 24	StegoGAN [26]	0.81	26.05	43.60	0.029	0.79	26.28	62.59	0.065	0.49	17.91	90.15	0.017
Ours	ThermalMask	**0.88**	**30.17**	38.94	0.026	**0.82**	**27.17**	**46.61**	**0.030**	**0.57**	**19.07**	89.92	0.016

and semantic attention maps to 7 can achieve the best balance between computational efficiency and generation effectiveness.

D. Quantitative Evaluation

To quantify the performance of translation effects, this section trains the models on the training sets of three datasets of VI-to-IR conversion, and conducts inference on the test sets of each dataset. We trained and calculated four indicators—SSIM, PSNR, FID, and KID—for all compared methods, and performed quantitative comparisons among different methods.

The experimental results on the AVIID dataset are shown in Table V, with bold indicating optimal performance and underline indicating suboptimal performance. Overall, the model presented in this article achieved significant comprehensive performance on these datasets. On the three subdatasets of AVIID, ThermalMask achieved eight optimal performances and two suboptimal performances out of 12 metrics. Although the performance of our model is slightly inferior in the FID and KID indicators, it has achieved a leading effect in the SSIM indicator, with SSIM results of 0.88, 0.82, and 0.57, respectively, indicating that our model is accurate in mapping semantic structures. In contrast, CUT, PUT, and RGB-TIR methods achieved optimal performance on few of these metrics. Additionally, the calculation results on the IRVI and DroneCoast datasets are shown in Table VI. Our method achieved optimal performance on all indicators, with SSIM values of 0.68 and 0.78, and KID values of 0.061 and 0.035, respectively. Swin-UNIT and StegoGAN both achieved suboptimal performance in three metrics. Notable that although there is currently no calculation method that can directly measure the similarity of generated thermal IR features among the evaluation indicators, the SSIM, FID, and KID indicators used in this article can reflect the quality of generated IR images to some extent. The results indicate that the thermal feature-guided image generation method of ThermalMask is more suitable for cross-modal translation tasks from VI to IR images.

Furthermore, to assess the generalization performance of the model, we designed cross-dataset generalization experiments on real-world scenarios, with the results presented in Table VI. Specifically, the AVIID-1 dataset, collected during the day, served as the training data for the generative model, while the AVIID-2 dataset, collected at night, was used for inference. Then, the SSIM and FID were used to evaluate the effectiveness of generating IR images. These two subdatasets were collected under different conditions. Due to the significant

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TABLE VI

QUANTITATIVE COMPARISON ON IRVI AND DRONECOAST DATASETS. BOLD AND UNDERLINE DENOTE THE BEST AND SUBOPTIMAL PERFORMANCE, RESPECTIVELY

Public	Method	IRVI				DroneCoast				Generalized	
		SSIM↑	PSNR↑	FID↓	KID↓	SSIM↑	PSNR↑	FID↓	KID↓	SSIM↑	FID↓
ICCV 17	CycleGAN [12]	0.56	18.52	110.66	0.120	0.68	24.24	170.90	0.166	0.28	270.58
ECCV 20	CUT [16]	0.63	20.77	89.96	0.084	0.63	22.90	141.04	0.125	0.27	247.72
MM 23	Swin-UNIT [25]	0.65	18.92	89.11	0.087	0.73	25.16	128.19	0.095	0.31	263.22
MM 22	PUT [18]	0.63	19.31	114.53	0.106	0.64	23.49	139.25	0.103	0.26	280.35
CVPR 22	HnegSRC [27]	0.57	18.57	131.08	0.149	0.70	22.85	154.47	0.099	0.27	251.19
ICRA 23	RGB-TIR [46]	0.62	19.98	133.79	0.140	0.71	24.72	176.66	0.081	0.28	246.85
TGRS 24	DR-AVIT [59]	0.65	18.37	114.06	0.114	0.56	18.19	198.93	0.119	0.22	317.51
TPAMI 24	DMT [38]	0.67	21.24	180.76	0.213	0.61	23.32	280.48	0.169	0.24	275.41
CVPR 24	StegoGAN [26]	0.67	21.68	97.80	0.103	0.60	20.76	136.92	0.040	0.25	261.60
Ours	ThermalMask	**0.68**	**22.24**	**75.32**	**0.061**	**0.78**	**25.31**	**122.97**	**0.035**	**0.34**	**239.03**

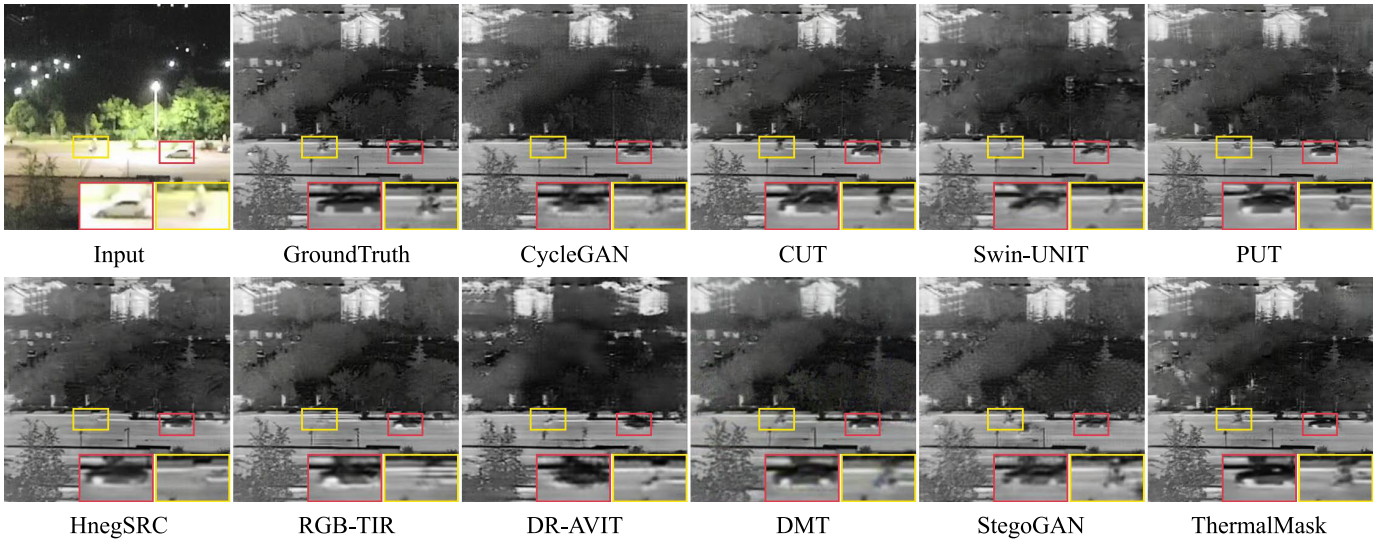


Fig. 8. Qualitative comparisons on the AVIID-2 dataset (night scene in drone remote sensing).

difference in feature distribution of IR data collected at night and collected during the day, the SSIM results of all methods have decreased notably. Nevertheless, the results of ThermalMask still exhibited significant advantages, indicating that our method has generalization ability to some extent. By reconstructing the texture and edges of images with high-frequency information, the model remains attentive to detailed features in the image, even on test data never seen before. Therefore, refining the generated features can make the model more effectively generalize to new datasets.

We conducted inference on VI images with input pixels of 256 × 256 on the IRVI dataset to evaluate the computational efficiency and inference time of various methods, to quantify their practical deployment efficiency, the results are shown in Table VII. ThermalMask has shown good performance in terms of computational efficiency, with only 11.88 M parameters and 70.26G FLOPs. Compared to other methods, the number of parameters and computational complexity of this method remain at a relatively low level. Specifically, the number of parameters for ThermalMask is very similar to those of CycleGAN, CUT, PUT, HnegSRC, and StegoGAN, and the number of FLOPs is slightly higher than these advanced methods. This increase is due to the introduction of the SMGN and SAGN in the model. However, the number of parameters in DR-AVIT is 65.04 M, and the number of FLOPs has significantly increased to 433.46G, indicating its relatively low computational efficiency. The DMT method reaches 398.10 M parameters, and due to the preset number of iterations in its diffusion denoising process, the number of FLOPs exceeds 293.65G. Noting that Swin-UNIT achieved the best results in terms of both parameter number and floating-point computation. This is because it introduces the Swin Transformer model as a feature extraction network, achieving a lightweight design. Moreover, we obtained the inference time on the entire dataset and calculated the average inference time for a single image. The inference time of ThermalMask for a 256 × 256 pixel image is 48.3 ms, slightly higher than CUT (45.2 ms) and CycleGAN (46.7 ms), and the lightweight model Swin-UNIT (32.6 ms), the inference time of ThermalMask is still within an acceptable range and has an advantage in the performance. Overall, the proposed ThermalMask has moderate computational complexity and parameter scale, achieving inference time similar to mainstream methods and demonstrating good computational efficiency. Its network design achieves an effective balance between overall performance and resource consumption, suitable for scenarios

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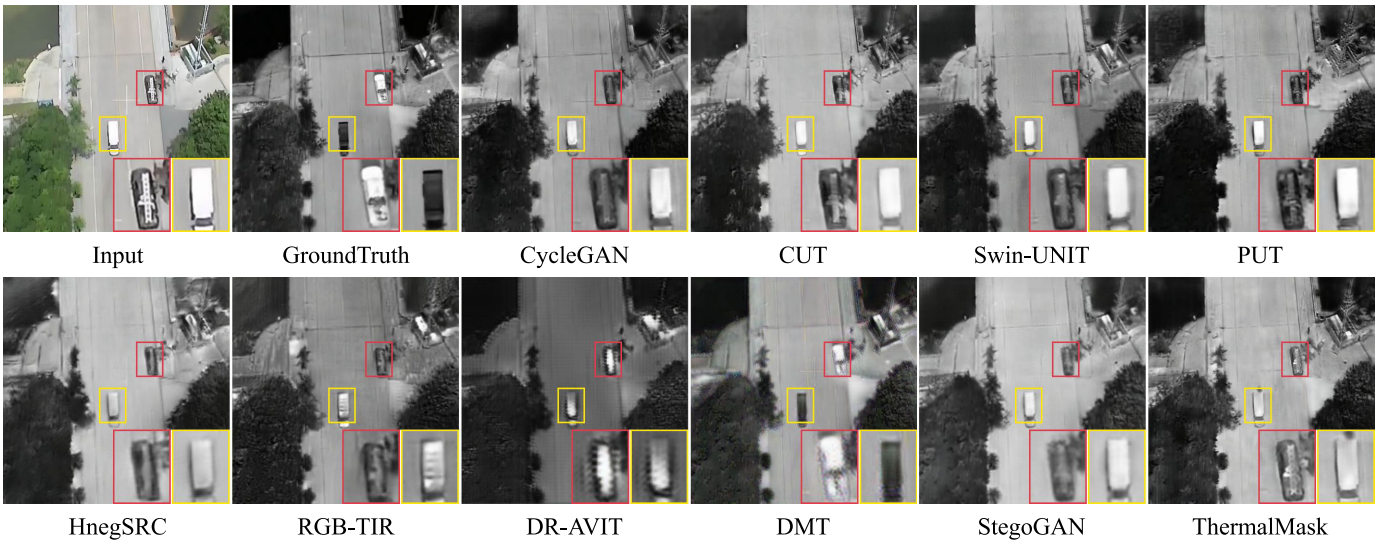


Fig. 9. Qualitative comparisons on the AVIID-3 dataset (road scene in drone remote sensing).

TABLE VII
COMPARISON OF COMPUTATIONAL EFFICIENCY OF DIFFERENT METHODS

Method	Parameter (M)	FLOPs (G)	Inference Time (ms)
CycleGAN [12]	11.38	64.13	46.7
CUT [16]	11.38	64.13	45.2
Swin-UNIT [25]	3.35	19.84	32.6
PUT [18]	11.38	64.13	45.9
HnegSRC [27]	11.38	64.13	41.0
RGB-TIR [46]	30.05	465.94	56.3
DR-AVIT [59]	65.04	433.46	68.5
DMT [38]	398.10	>293.65	117.1
StegoGAN [26]	11.38	64.11	44.7
ThermalMask (Ours)	11.88	70.26	48.3

where computing resources are limited but certain performance requirements still need to be guaranteed. In the future, we will further explore model compression or hardware acceleration technologies to further improve the inference speed of the model.

E. Visual Qualitative Comparison

The visualization of inference results on three datasets are shown in Figs. 7–11. It can be seen that, although CycleGAN, Swin-UNIT, and HnegSRC methods can convert from the VI domain to a similar IR domain, they excessively preserve the visual features of the source VI image during the conversion, and fail to fully follow the unique rules of IR images. For example, a running vehicle should exhibit bright features due to thermal radiation, while CycleGAN preserves too much of the vehicle's white appearance. Swin-UNIT and HnegSRC mistakenly convert its overall color tone to gray, deviating from the true representation of heat distribution in real IR imaging. The CUT method shows good performance in low-resolution traffic scenes, such as displaying the thermal characteristics of vehicles in Figs. 7 and 11. But when dealing with high-resolution aerial coastal scenes, the model exhibits ghosting. The root cause of this phenomenon may lie in the insufficient contrast constraints applied by a single generator, which cannot effectively handle complex details and texture information in high-resolution images. In most cases, the PUT method may find it difficult to focus on IR thermal feature areas, resulting in distortion of foreground objects in generated IR images. This indicates that PUT is more suitable for creative tasks such as style transfer. The RGB-TIR method can capture foreground objects and thermal features in images, but the generated IR features are not precise enough, thus further refinement of the boundary positions of the IR features is still needed. The DR-AVID method has diverse results, making it easy to change the contour and structure of foreground objects in generated images, such as the deformation of vehicles in Figs. 8 and 9. The DMT method is a translation method based on the diffusion model. When directly applied to IR image generation, it may result in weak color traces in the generated image. This is due to the inherent characteristic of the diffusion model gradually recovering data from noise. StegoGAN can accurately reconstruct the foreground and background contours of an image, but the location of the highlighted areas generated in Fig. 10 is not accurate enough.

From the image, it can be observed that the ThermalMask method produces the most realistic IR thermal features while preserving as much detail as possible in VI images. The visual results indicate that the thermal feature-guided generation and feature refinement proposed in this article yield good results in VI-to-IR image translation tasks.

F. Object Detection Extension Experiment

To explore the performance of VI-to-IR image translation in practical applications, this section introduces IR image data generated by various translation models into downstream object detection tasks, using the AVIID-1 dataset for research. Specifically, the current advanced Yolov9 [68] object detection model is selected as the evaluation benchmark to indirectly assess the quality of the translated image. This is done by comparing the impact of different training data sources on detection performance. In this section, a strict distinction is

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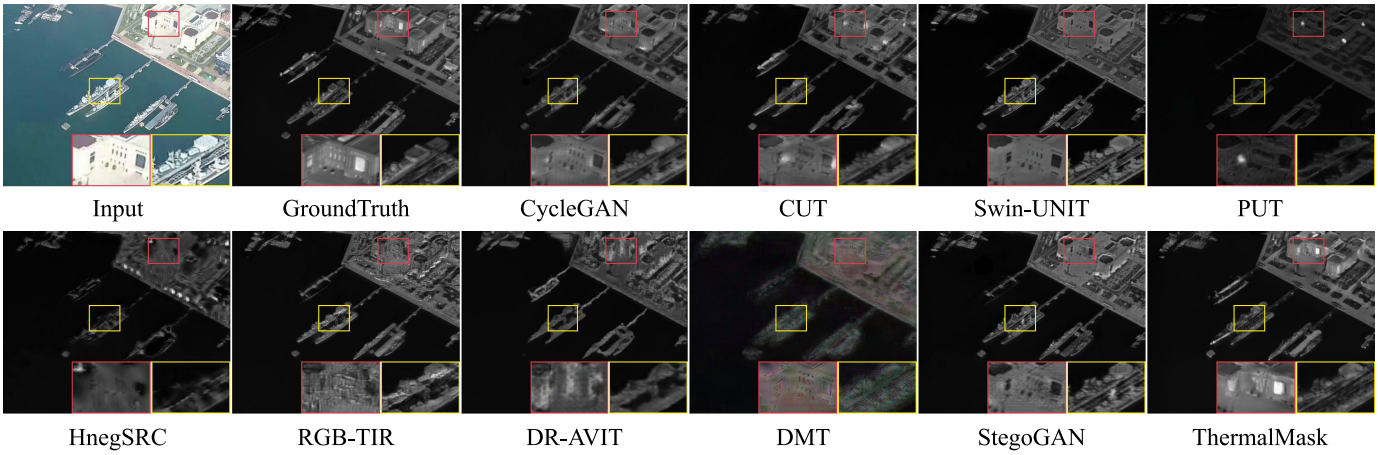


Fig. 10. Qualitative comparisons on the DroneCoast dataset (coastal scene in drone remote sensing).

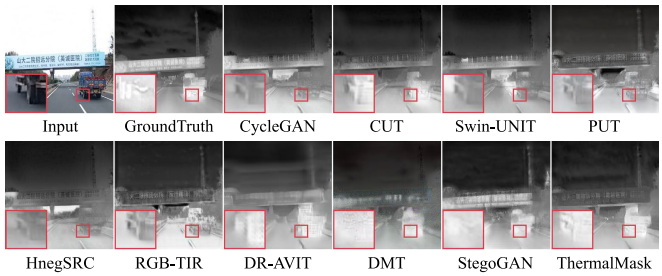


Fig. 11. Qualitative comparisons on the IRVI dataset (driving scene).

made between the training set and the test set. The IR images generated by the translation model are used as the training data for the object detection model, while the real IR images are used as the test set to evaluate the benefit of cross-model generated images for downstream tasks. Additionally, two control experiments were conducted: one group was trained directly using VI images, and the other group was trained using real IR images. The test indicators in Fig. 12 and the visualization results in Fig. 13 show that the object detection model trained directly using VI images has significantly poorer detection performance in the IR domain. This further confirms the challenge associated with cross-modal data applications.

Analysis found that when using IR images generated by ThermalMask as training data, the performance of object detection was significantly improved, achieving results comparable to those obtained by training directly using real IR data. This method ranked first among these comparative methods, while StegoGAN and Swin-UNIT achieved suboptimal results. The results validated the effectiveness of the IR images generated by the ThermalMask model for downstream task applications, indicating a high degree of similarity between the generated images and real IR images at the feature level. Although the performance of images generated by ThermalMask in object detection tasks is slightly inferior to that of training directly with real IR images, this does not prevent it from being an efficient data augmentation method, especially in scenarios where real IR data is scarce or difficult to obtain. More importantly, the result provides strong evidence that high-quality generated images can effectively enhance the performance of downstream tasks, offering new

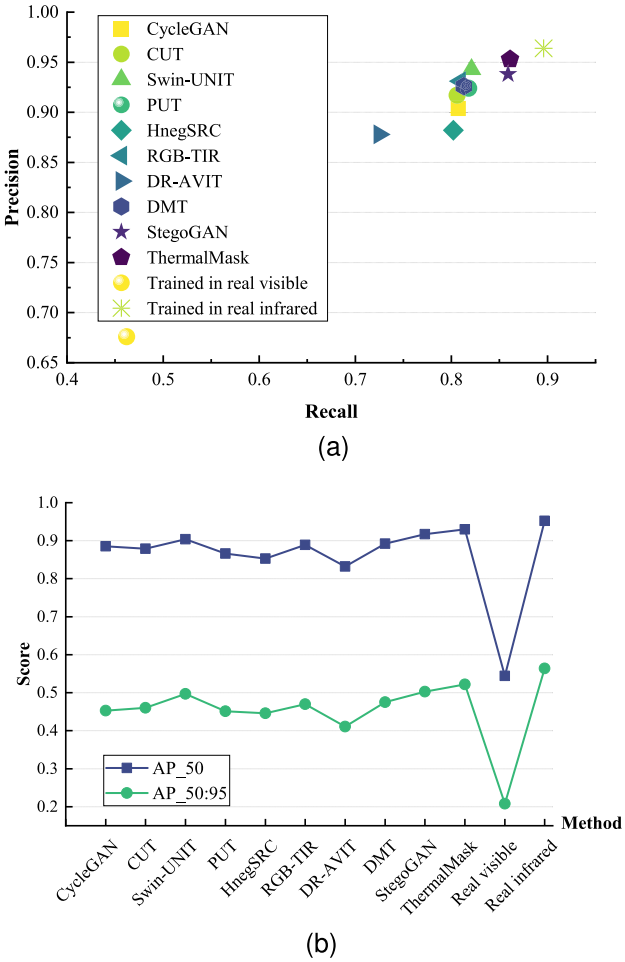


Fig. 12. Test results of object detection metrics using different training data. (a) Precision and recall. (b) AP_50 and AP_50:95.

insights for future research on cross-modal learning of VI-to-IR images and the scientific utilization of data.

G. Discussion

While our proposed model demonstrates efficacy in IR image generation, its applicability may be constrained by potential limitation stemming from the robustness of remote sensing training datasets. Remote sensing data is often affected

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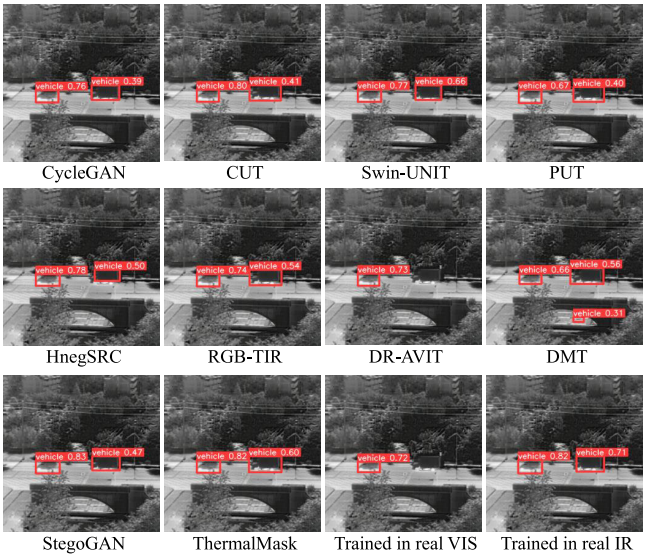


Fig. 13. Applying training data from different generation sources to object detection.

by degradation, noise, and spectral variability during the imaging process, which can have a significant impact on the quality of the collected VI or IR images. When faced with such changes, ALMM [69] models the spectral changes caused by lighting or brightness changes using endmember dictionaries, and then introduces a spectral change dictionary to simulate other spectral changes caused by environment conditions, instrument configurations, and material nonlinear mixing effects. This physical modeling scheme provides great inspiration for the above challenges faced in image generation.

The proposed ThermalMask enhances result robustness through two key characteristics. The first is the design of cross-modal learning network. Spectral changes are mainly caused by environmental factors such as lighting and temperature, as well as sensor-specific noise, which can distort the distribution of thermal features in IR images. ThermalMask does not rely on single mode learning, and its search for potential thermal feature areas is conducted by masking attention with salient features in VI images. Although saliency features are influenced by lighting conditions, they are insensitive to temperature changes and sensor-specific noise, making the model produce robustness to spectral changes. The second is the spatial–spectral joint constraint mechanism in ThermalMask model. It can suppress the frequency band dominated by noise by learning the frequency domain mask, while preserving the mid- to high-frequency signals related to the scene structure. The discriminator actively adjusts the frequency domain feature distribution of the output image to avoid factors such as noise introduced by the system by minimizing the spatial–spectral adversarial loss. However, the optimality and applicability in this situation is currently unexplored. Therefore, how to further conquer this problem will be an interesting topic for future research.

V. Conclusion

This article proposes a new cross-modal feature learning and image translation method for remote sensing data, converting VI images into IR ones that align with thermal features, thus alleviating the scarcity of IR data for subsequent tasks. We have designed a customized framework, ThermalMask, which integrates saliency feature learning and feature refinement constraints into the generative network. Specifically, an SMGN was designed within the generator to produce foreground and background masks, and an SAGN was also designed to generate attention maps. Additionally, a new constraint was specifically designed to refine the generation of IR features, combining pixel space and frequency spectral space to ensure that the generative network focuses more on the expressing of IR features. The experimental results on several challenging datasets demonstrate that our method outperforms current state-of-the-art general image translation methods and VI-to-IR image conversion methods in both qualitative and quantitative evaluations. The results confirm the effectiveness of thermal feature and multidomain constraints for cross-modal IR image generation, as well as the feasibility of applying the generated data to downstream tasks. Future research can further explore the diffusion-based IR generation models to enhance robustness against imaging noise and spectral uncertainty. Additionally, the compression or lightweight for these translation models is also a potential research.

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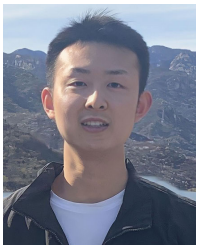
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